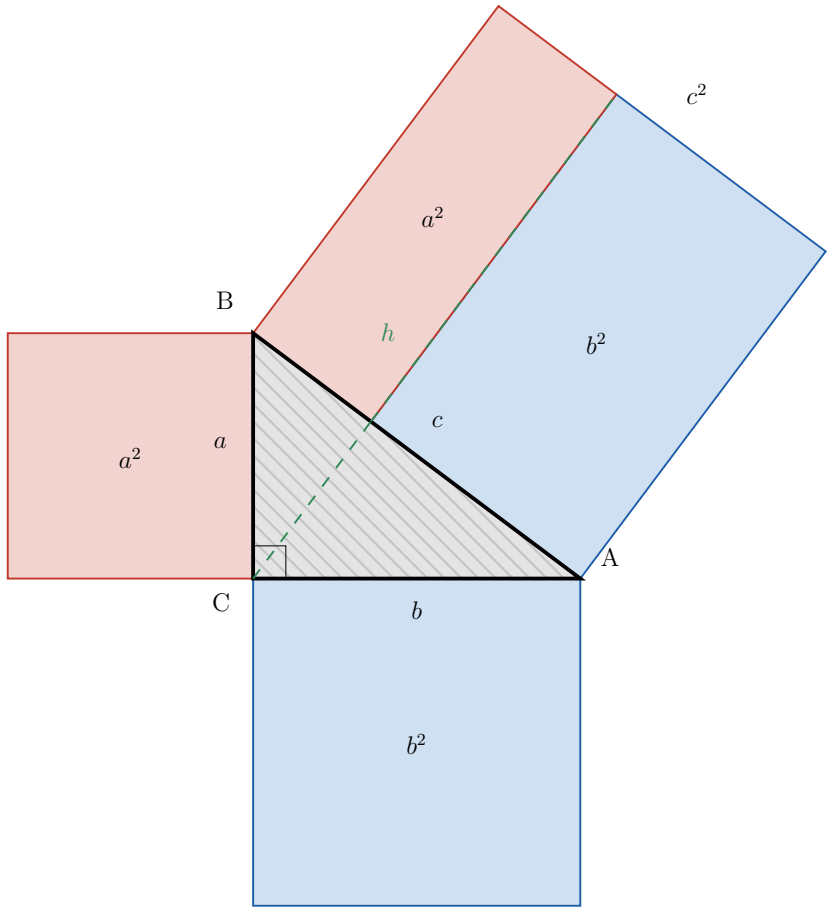
Pythagorean theorem: Euclid's figure on a 3-4-5 triangle; the altitude splits the hypotenuse square into rectangles equal to $a^2$ and $b^2$.

## Pythagorean theorem

Euclid's figure: the squares on the legs equal the square on the hypotenuse




$$a^2 + b^2 = c^2$$

Areas for $a = 3$, $b = 4$, $c = 5$:

- $a^2 = 9$
- $b^2 = 16$
- $c^2 = 25$

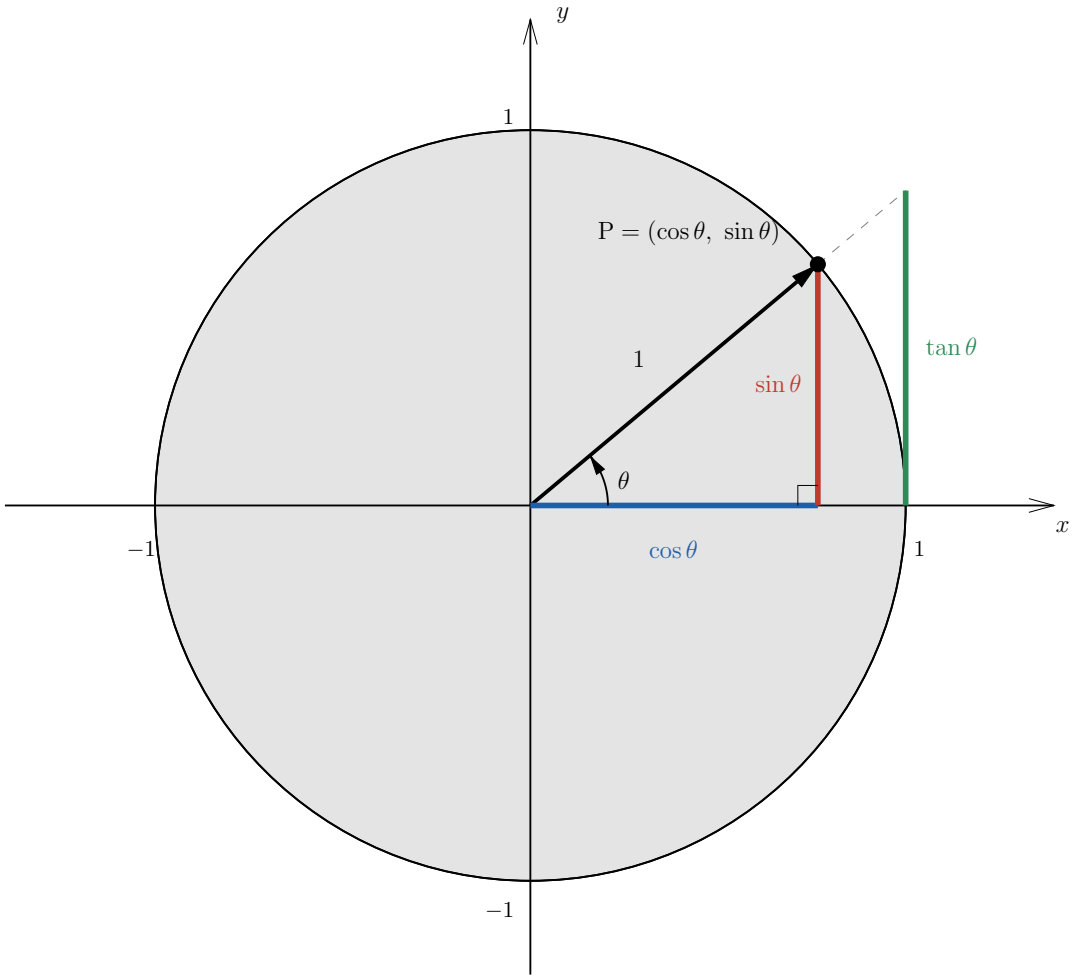
$9 + 16 = 25$

The altitude $h$ from the right angle cuts the big square in two rectangles: blue $= b^2$, red $= a^2$.

Unit circle: $\sin\theta$, $\cos\theta$ and $\tan\theta$ as segment lengths for $\theta = 40°$.

## Unit circle and the trigonometric functions

$\sin\theta$, $\cos\theta$ and $\tan\theta$ as lengths of segments, here for $\theta = 40°$




$\sin^2\theta + \cos^2\theta = 1$

$\tan\theta = \sin\theta / \cos\theta$

For $\theta = 40°$:

$\cos\theta = 0.766$

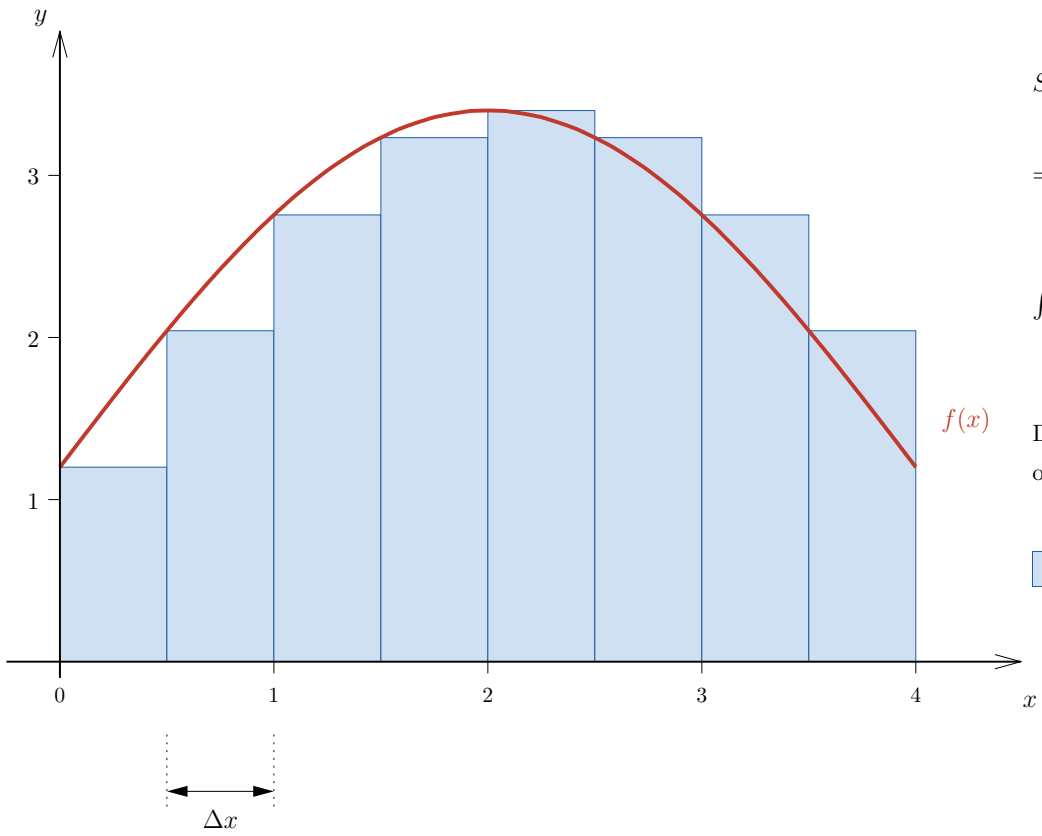
$\sin\theta = 0.643$

$\tan\theta = 0.839$

Riemann sum: left-endpoint sum with 8 rectangles under $f(x) = 1.2 + 2.2\sin(\pi x/4)$ on $[0, 4]$.

## Riemann sum

Left-endpoint sum with 8 rectangles approximating a definite integral




$S_8 = \sum_i f(x_i)\,\Delta x$

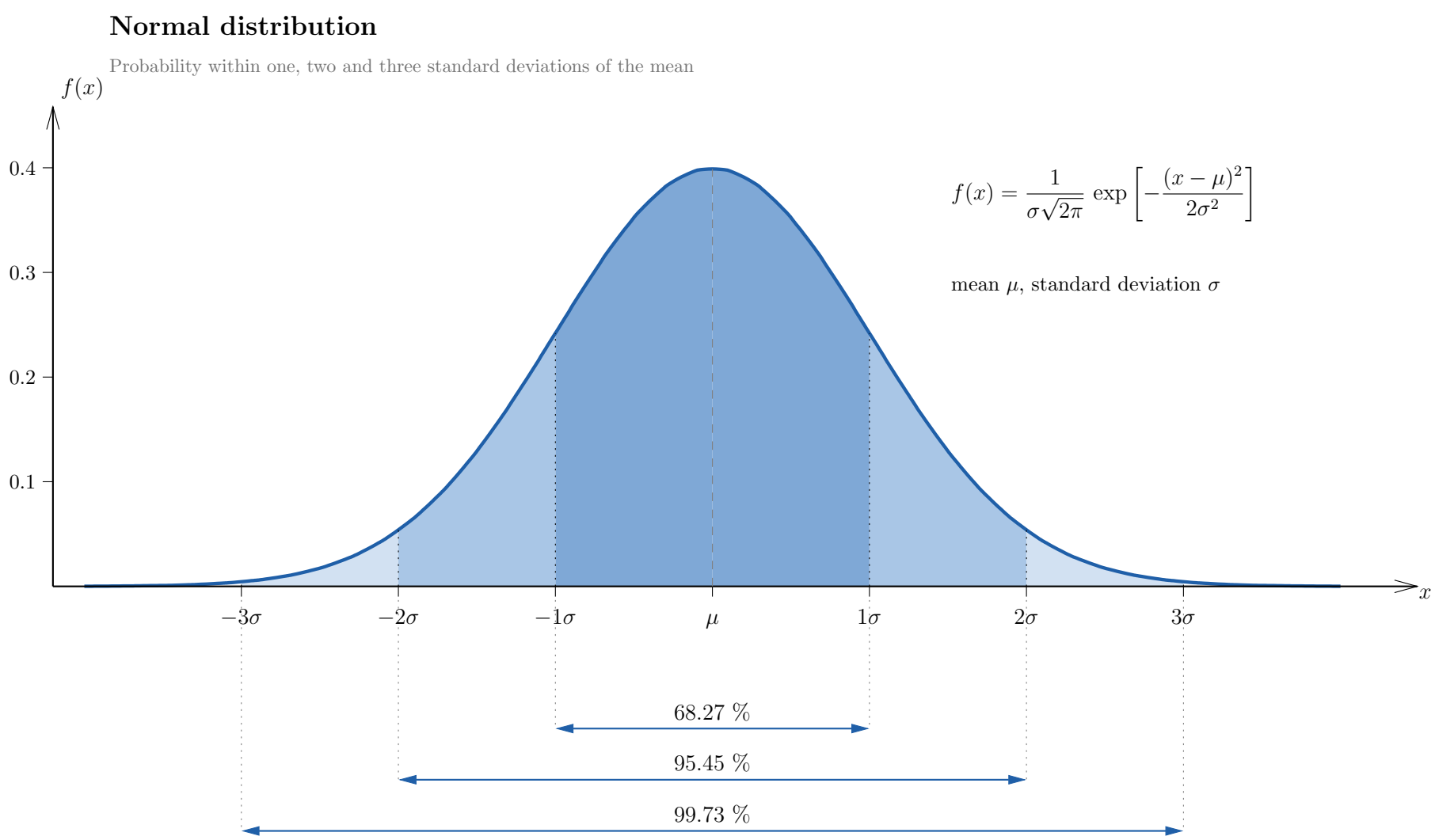
$= 10.330$

$\int_0^4 f(x)\,dx = 10.402$

Doubling n halves the error of the left-endpoint rule.

rectangle $f(x_i)$

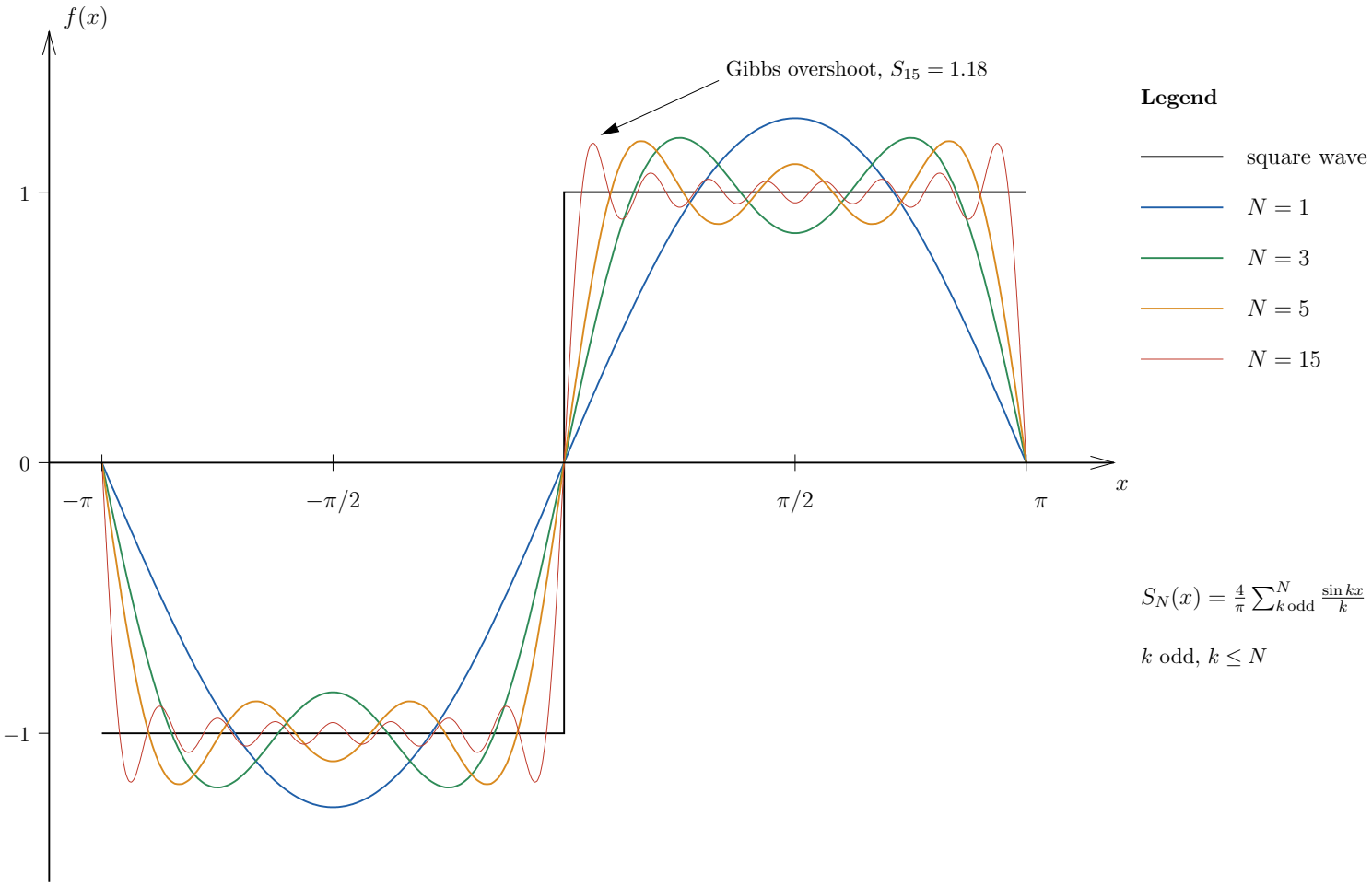
Normal distribution with the 68.27, 95.45 and 99.73 % regions of $\pm 1, 2, 3\sigma$.



Fourier series of a square wave: partial sums $N = 1, 3, 5, 15$ and the Gibbs overshoot.

## Fourier series of a square wave

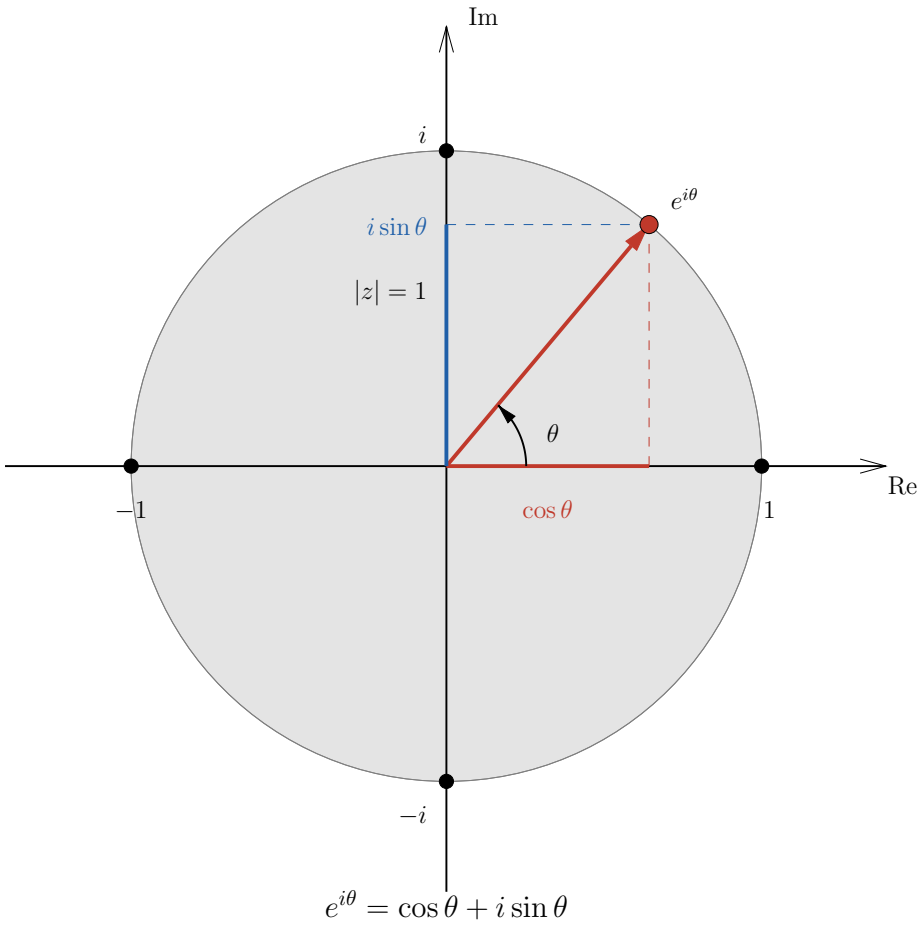
Partial sums with 1, 3, 5 and 15 terms; the overshoot at the jump does not vanish (Gibbs)



Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ on the unit circle of the complex plane.

## Euler's formula in the complex plane

$e^{i\theta}$ runs around the unit circle; its projections are $\cos\theta$ and $\sin\theta$



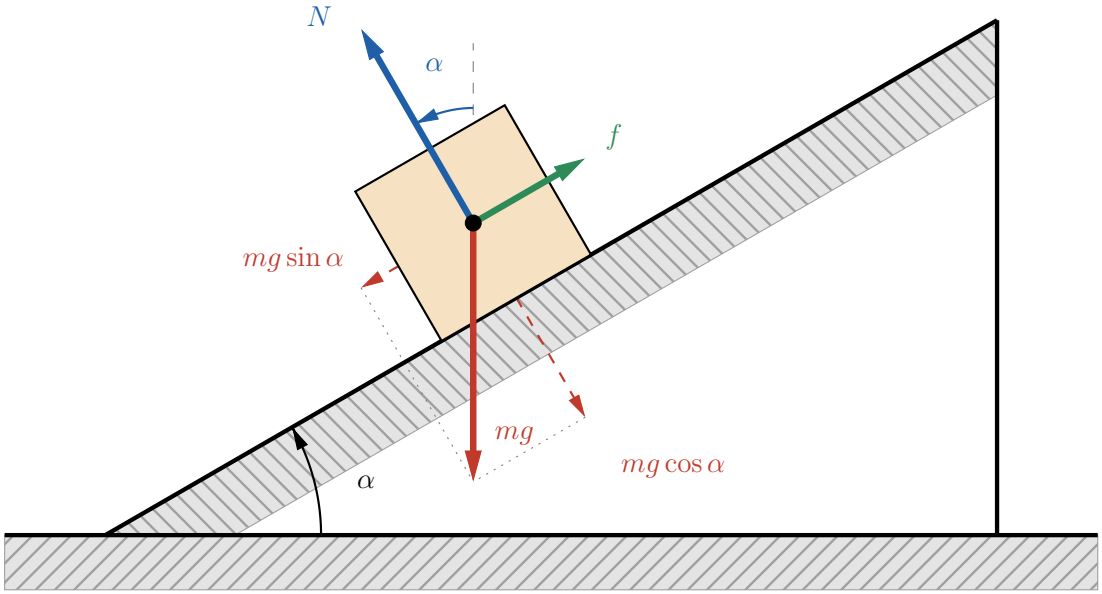

Euler's identity:

$e^{i\pi} + 1 = 0$

Free-body diagram of a block at rest on a 30° incline: $mg$, $N$, $f$ and the weight components.

## Block on an inclined plane

Free-body diagram of a block at rest on a 30° incline




At rest:

$N = mg\cos\alpha$

$f = mg\sin\alpha$

$f \le \mu_s N$

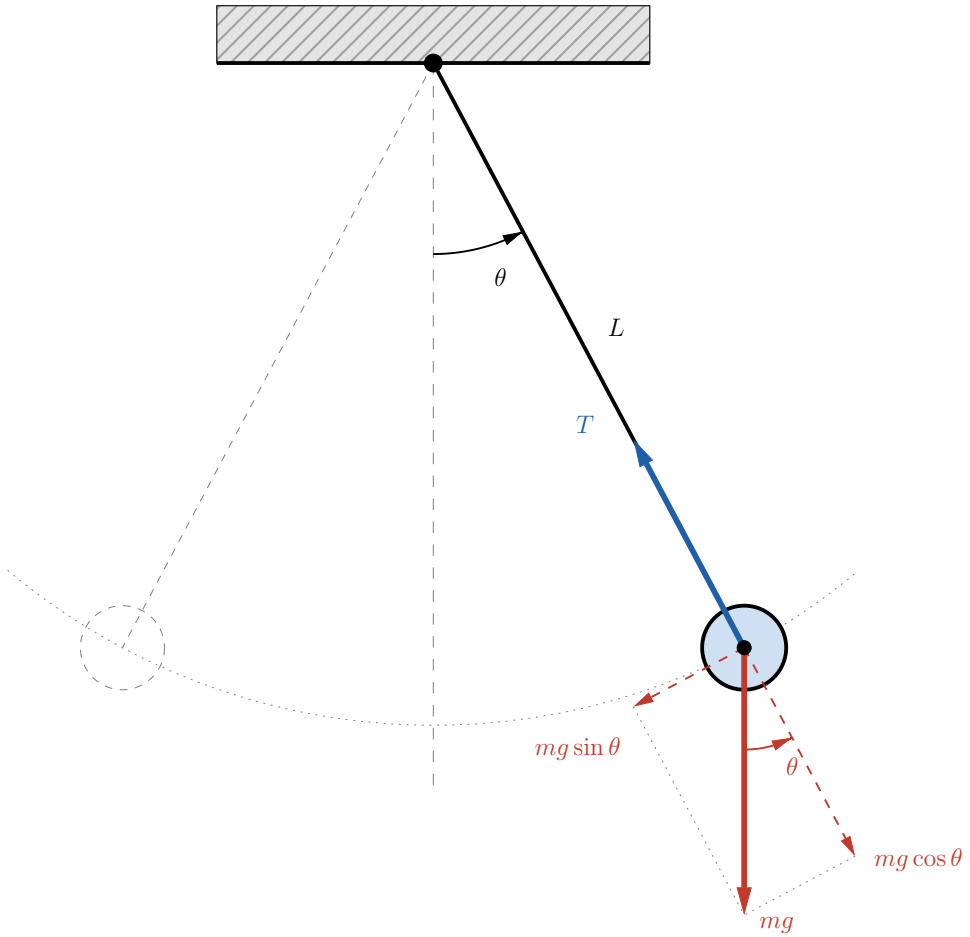
$\Rightarrow \tan\alpha \le \mu_s$

$\mu_s$: coefficient of static friction

Simple pendulum at a turning point: weight, tension and the components of $mg$.

## Simple pendulum

Forces on the bob at the turning point of its swing




At the turning point ($v = 0$):

$T = mg\cos\theta$

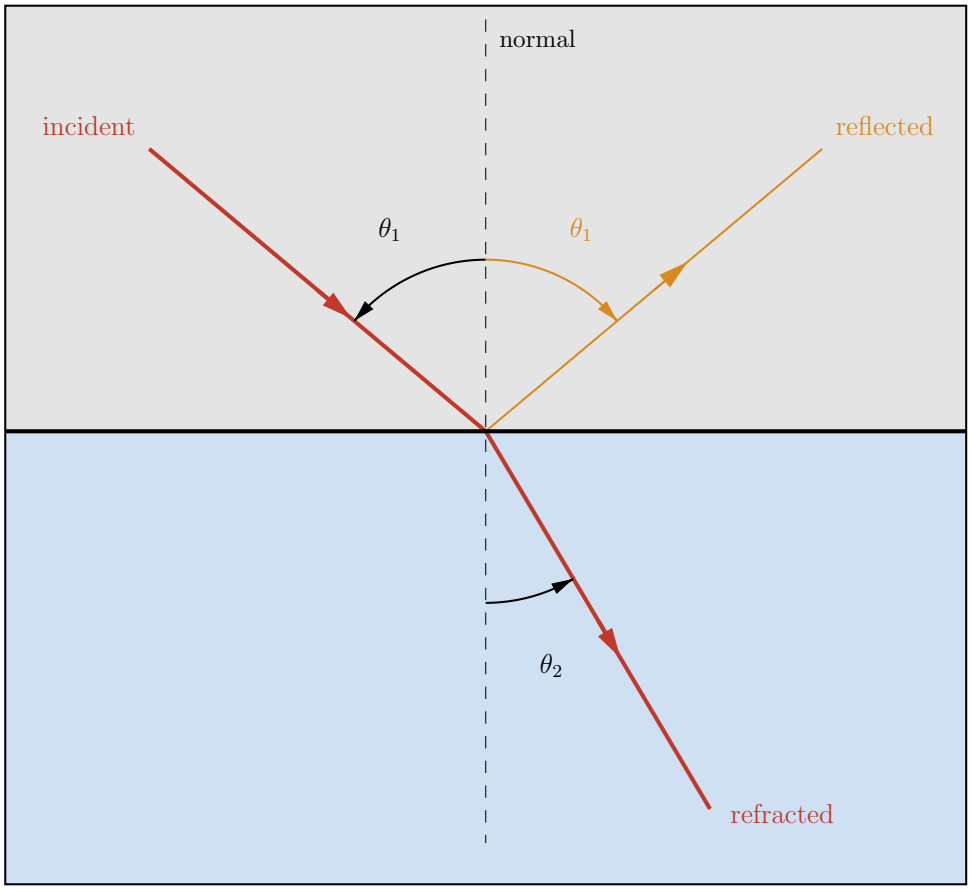
$mL\,\dfrac{d^2\theta}{dt^2} = -mg\sin\theta$

Small angles: $\omega = \sqrt{g/L}$

Snell's law: refraction from air ($n_1 = 1.0$) into glass ($n_2 = 1.5$), $\theta_1 = 50°$, $\theta_2 = 30.7°$.

## Refraction: Snell's law

A ray entering the denser medium bends toward the normal




$n_1 = 1.00$ (air)

$n_1 \sin\theta_1 = n_2 \sin\theta_2$

$\sin\theta_2 = \sin 50°/1.50 = 0.511$

$\theta_2 = 30.7° < \theta_1 = 50°$

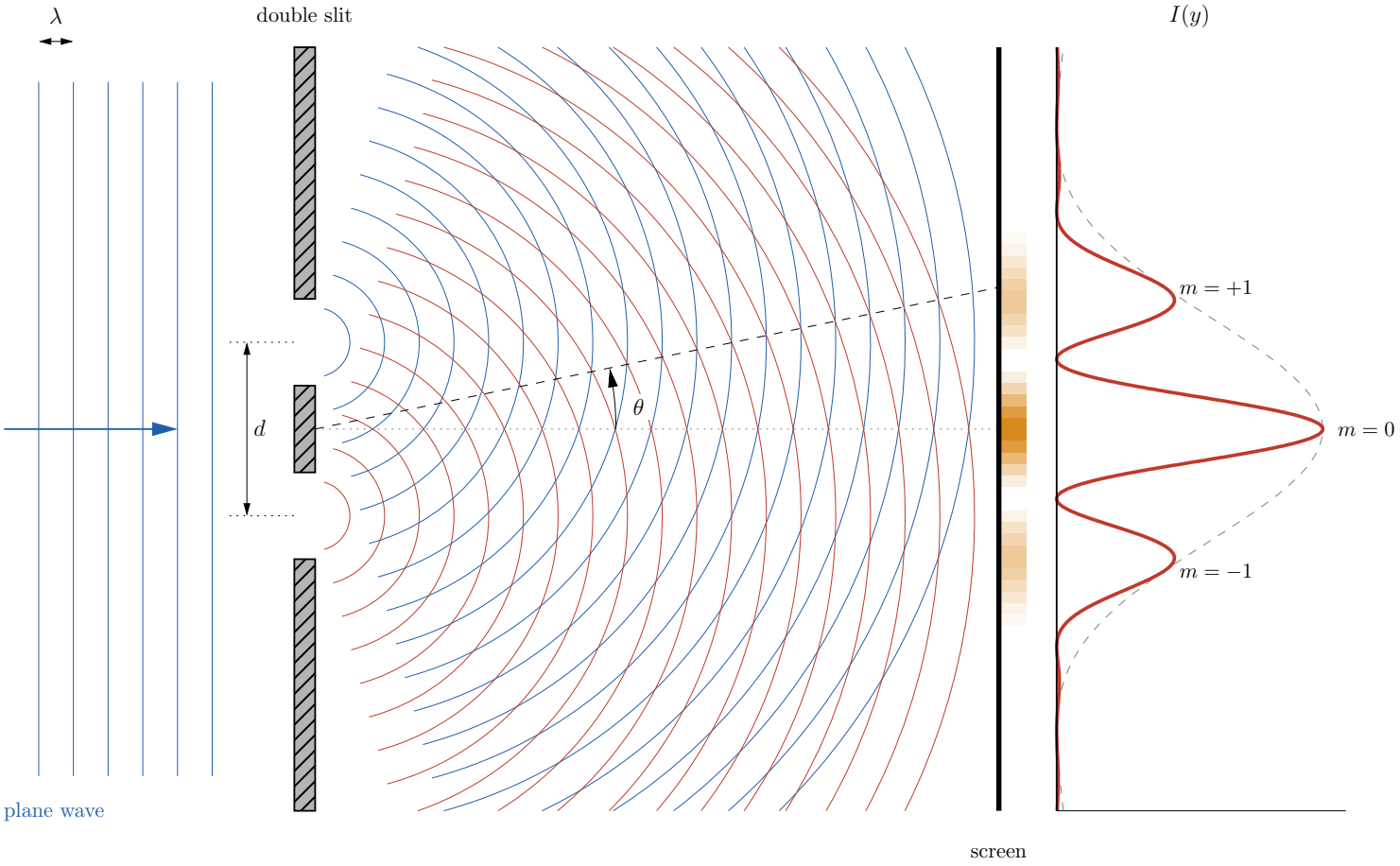
$n_2 > n_1$: the ray bends toward the normal.

$n_2 = 1.50$ (glass)

Young's double slit: wavefronts, screen fringes and the $\cos^2 \times \mathrm{sinc}^2$ intensity pattern.

## Young's double-slit experiment

Interference of two coherent wavefronts: $\cos^2$ fringes under a single-slit $\mathrm{sinc}^2$ envelope

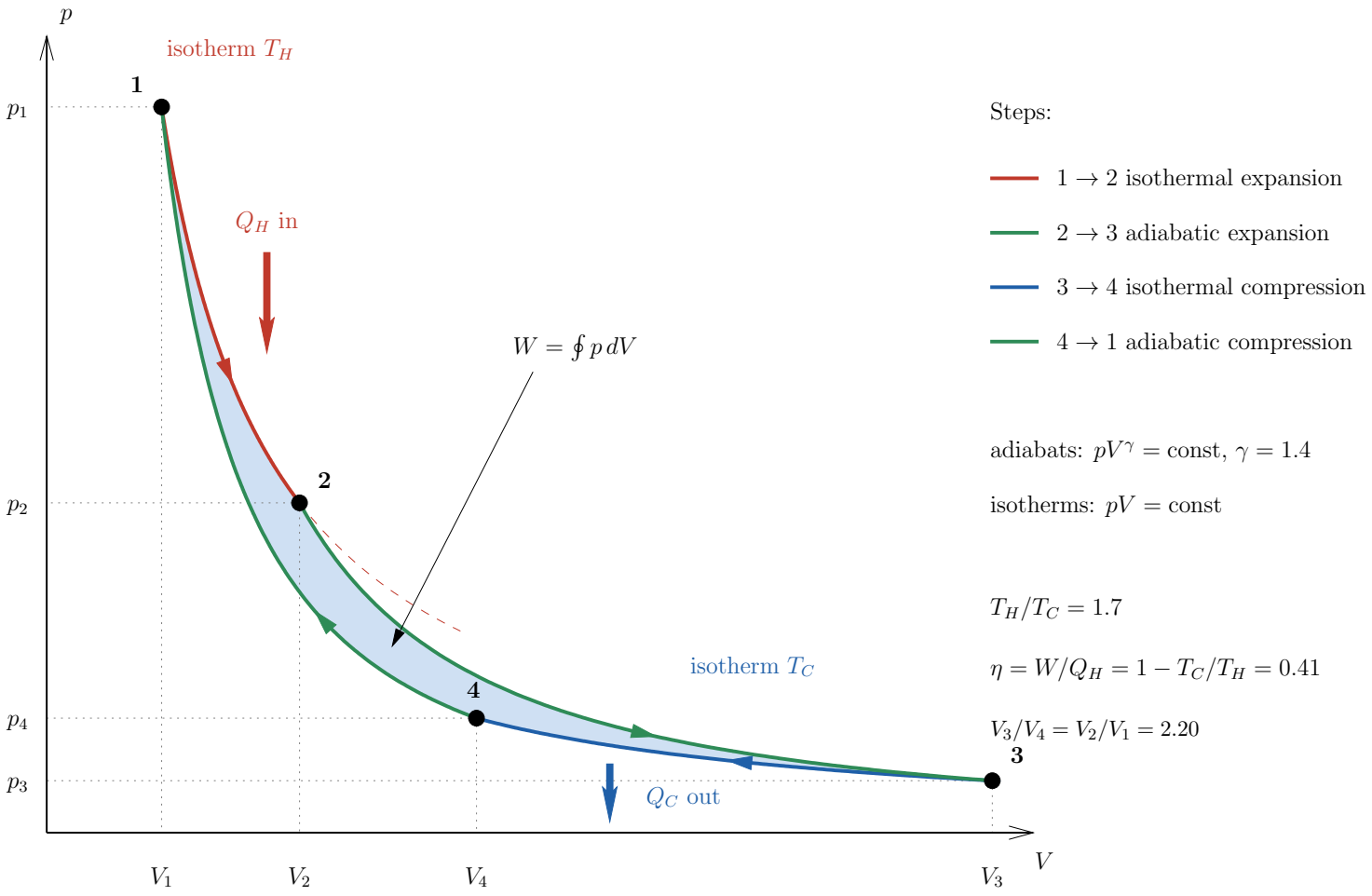



$d \sin\theta = m\lambda$ (maxima), $d = 5\lambda$, $a = d/2$: orders $\pm 2$ are missing (single-slit zeros)

Carnot cycle in the $p$–$V$ plane: two isotherms and two adiabats ($\gamma = 1.4$), $T_H/T_C = 1.7$.

## Carnot cycle

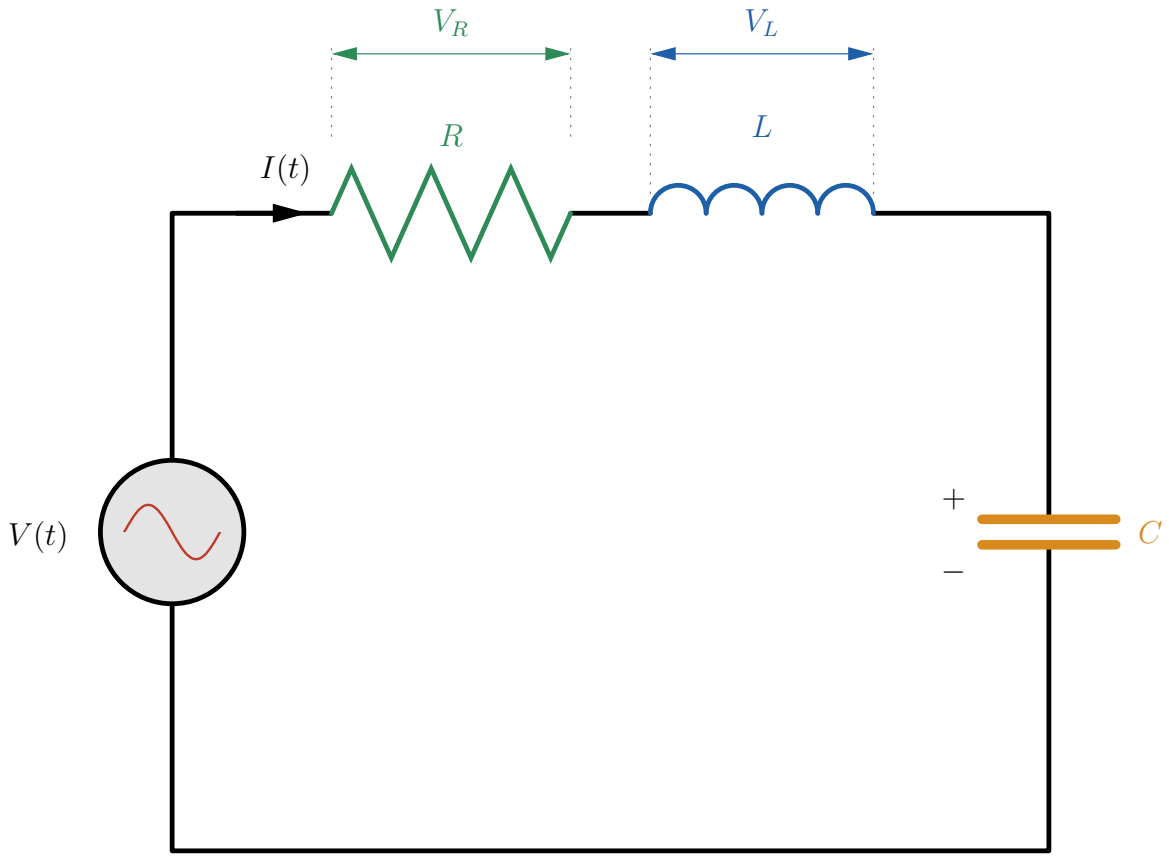
Two isotherms and two adiabats ($\gamma = 1.4$); the enclosed area is the net work $W$



Series RLC circuit with AC source, current and voltage drops.

# Series RLC circuit

AC source driving a resistor, an inductor and a capacitor in series




Kirchhoff's loop rule:

$$L\frac{dI}{dt} + RI + \frac{Q}{C} = V_0 \cos \omega t$$

$$\omega_0 = 1/\sqrt{LC}$$

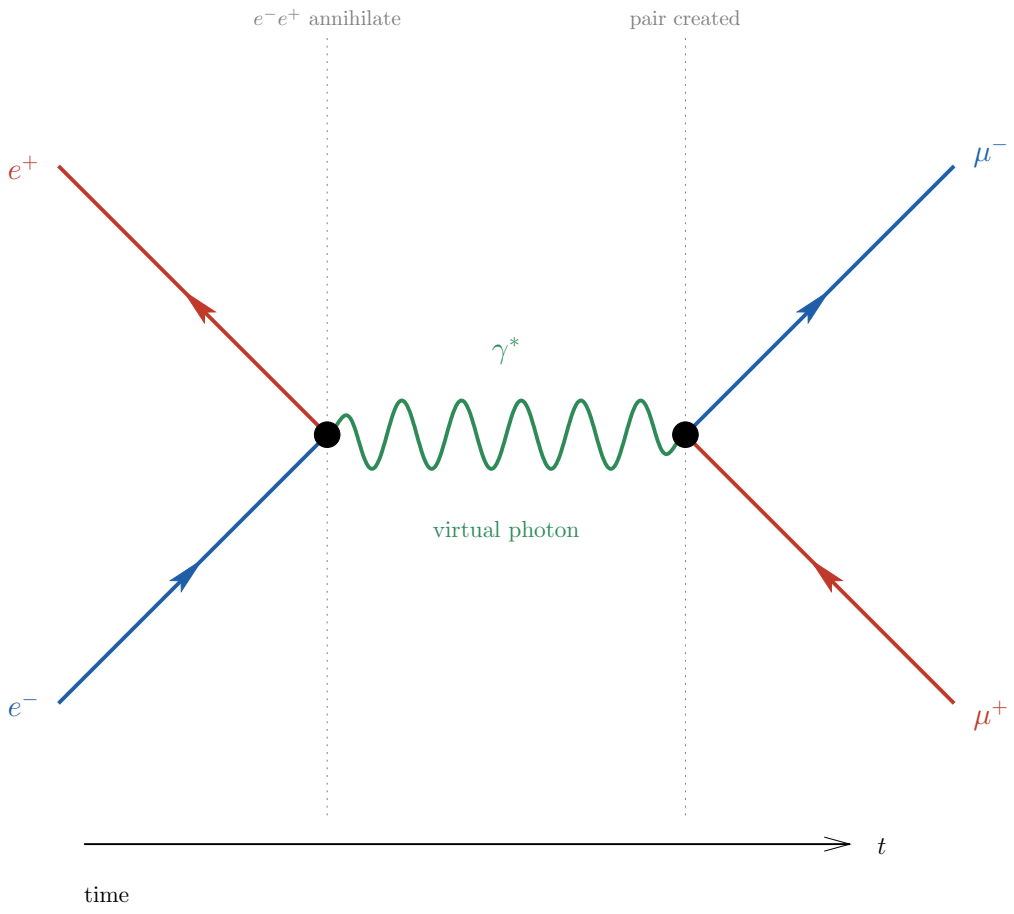
resonance frequency

$$|Z| = \sqrt{R^2 + (\omega L - 1/\omega C)^2}$$

Feynman diagram $e^- e^+ \to \gamma^* \to \mu^- \mu^+$, time to the right.

## Electron-positron annihilation into muons

Lowest-order QED process, time flows to the right




Process

$e^- e^+ \to \gamma^* \to \mu^- \mu^+$

Arrow: flow of lepton number; the antiparticle's arrow points backwards in time.

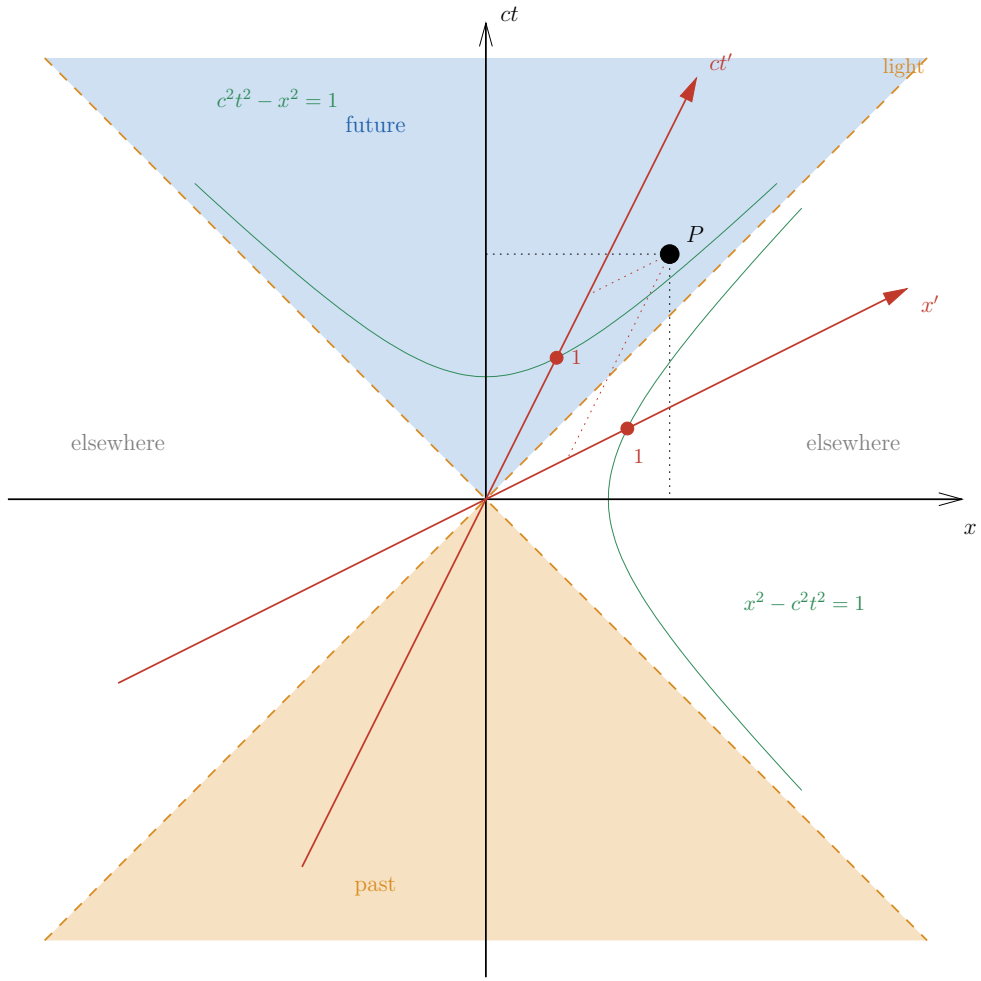
Cross section (QED):

$\sigma = 4\pi\alpha^2/3s$

Minkowski diagram: light cone and the axes of a frame moving at $v = 0.5\,c$, with calibration hyperbolas.

## Minkowski diagram

Light cone and the axes of a frame moving at $v = 0.5\,c$; the same event has coordinates in both frames




Frame $S'$ moves at $v = 0.5\,c$

$\beta = v/c = 0.5,\ \gamma = 1.155$

$ct'$ axis: $x = \beta\, ct$ (tilt 26.57°)

$x'$ axis: $ct = \beta x$

Event $P = (x, ct) = (1.5, 2.0)$

in $S'$: $x' = 0.58,\ ct' = 1.44$

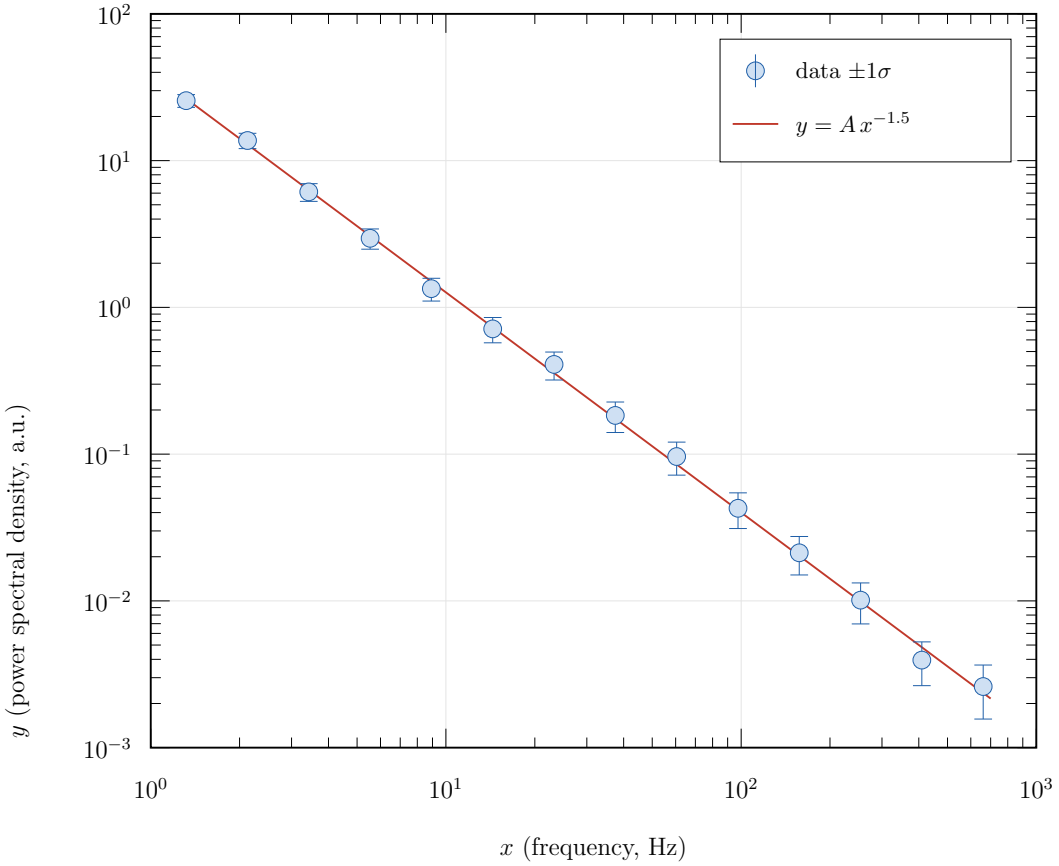
$(ct')^2 - x'^2 = (ct)^2 - x^2$

invariant interval: 1.75

Power law on log-log axes: data with error bars, fitted line and legend.

## Power law on log-log axes

Data with error bars and a least-squares line of slope $-1.5$



Damped harmonic oscillator $x(t)$ inside its exponential envelope.

## Damped harmonic oscillator

Underdamped motion $x(t)$ inside its exponential envelope

